Matias Freire

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EDUCATION

Stevens Institute of Technology — **Hoboken, NJ**
Bachelor of Science in Computer Science — **Sept 2024 – May 2028**

Relevant Courses**:** Algorithms, Data Structures, Discrete Structures, Computer Architecture and Organization, Multivariable Calculus, Vectors and Matrices, Probability, Mechanics, Biology, and Biotechnology

Certifications: Hugging Face AI Agents Fundamentals, Bloomberg Market Concepts

Awards: STEP Alumni Scholarship: Selected as 1 of 3 recipients out of 200+ members for outstanding leadership

EXPERIENCE

SHIFT Research Assistant — **Jan 2026 – Present**
Hanlon Financial Systems Center, SIT — Hoboken, NJ

- Selected for the SHIFT Research Assistant program; currently onboarding and preparing to support research initiatives in financial systems and quantitative analysis.

Deep Learning Research Assistant/Intern — **Aug 2025 – Present**
Advanced Quantum Materials Lab, SIT — Hoboken, NJ

- Studying preprocessing, synthetic data generation, and deep learning methods for MoS_2 defect characterization while designing, building, and maintaining a Dash–Plotly platform from scratch that integrates open-source pipelines for vacancy and polymorph defect analysis.
- Synthesized and presented nanomaterials research to Dr. Eui-Hyeok Yang and lab members through bi-weekly presentations, improving team-wide understanding of material properties and defect analysis methodologies.

Machine Learning Ignite Fellow — Financial Track — **Aug 2025–Jan 2026**
AI4ALL — Remote

- Completed an intensive applied machine learning fellowship, implementing linear regression and tree-based models using pandas and scikit-learn, with hands-on data preprocessing, feature engineering, and exploratory analysis on structured datasets.
- Collaborated on planning, developing, and presenting a capstone machine learning project, applying project management practices and ethical AI frameworks, with deployment and documentation in Jupyter Notebook and scikit-learn.

ACADEMIC PROJECTS

MoS2 Image Synthesis & Analysis Platform | PyTorch, Dash-Plotly | Github-Repo

- Designed and built from scratch an end-to-end Dash/Plotly research platform for deep learning–based STEM MoS_2 defect analysis, automating synthetic image generation with configurable defects and realistic noise injection, and integrating a ResUNet model for vacancy and polymorph classification

Colorstack Stevens Chapter Website | Next.js, React, Tailwind, Typescript | Github-Repo

- Built from scratch a modular, responsive Next.js / React website for the ColorStack Stevens chapter, serving 60+ members, with a reusable UI system established through rapid HTML/CSS prototyping and full implementation in TypeScript and Tailwind CSS.

Interactive-Neural-Network-Digit-Classifier | PyTorch, Streamlit, Numpy, Pygame | Github-Repo

- Designed, trained, and deployed a 4-layer CNN in PyTorch (96% MNIST accuracy), integrating a PyGame-based drawing interface and a Streamlit web app for real-time inference and visualization.

LEADERSHIP ROLES & PROFESSIONAL DEVELOPMENT

STEP Tutor-Counselor/Resident Assistant (TCRA) — **Jun 2025 – Aug 2025**
***Bridge Summer Program* 2025** — Hoboken, NJ

- Strengthened STEM readiness and academic success by leading evening Calculus I/II and Physics recitations for 50+ incoming freshmen and providing personalized Calculus tutoring to 25+ students weekly, including after-hours support, while serving as a Resident Assistant supporting residential life operations.

Head of Extern Relations & Co-Chair of Technology — **April 2025 – Present**
Colorstack Stevens Chapter — Hoboken, NJ

- Led growth and technical engagement initiatives, increasing active membership by 50% and online engagement by 25% through social media strategy and 14+ events per semester, and organizing peer-led sessions solving 32+ data structures problems per term.

SKILLS

Programming: Java, Python, C/C++, JavaScript
Web Tools: TypeScript, React, Tailwind, Next.js
Developer Tools: Git, GitHub, VS Code, Jupyter Notebooks
Libraries: Dash, Pandas, NumPy, PyTorch, Scikit-learn, Streamlit